

Advanced Landslide Detection Using Machine Learning and Satellite Imagery

The Threat of Landslides

- Landslides pose a significant global threat, causing loss of life and infrastructure damage.
- Statistics: Between 2004 and 2016, over 55,000 people died in 4,862 landslide events worldwide, with annual economic losses estimated at USD 20 billion (Geoenvironmental Disasters).
- Common in mountainous regions, landslides disrupt essential services like water and electricity (WHO Landslides).



Technological Solutions

- Early detection is critical for timely evacuation and disaster management.
- Satellite imagery enables large-scale monitoring of remote areas.
- Machine learning, particularly deep learning, automates detection with high accuracy.
- Your research leverages a U-Net model to process satellite data for pixel-wise landslide detection.

Related Work

Evolution of Landslide Detection

- Traditional Methods: Field surveys are labor-intensive and limited in scalability.
- Early Machine Learning: Support Vector Machines (SVMs) and Random Forests (RF) used for susceptibility mapping based on slope and rainfall.
- Deep Learning: Convolutional Neural Networks (CNNs) excel in spatial feature extraction from imagery.

Related Work

- **SVM Applications:** Studies like Koren et al. (2009) used SVMs for susceptibility mapping, effective for structured data but limited in spatial analysis.
- **CNN Advancements:** Zhang & Chen (2018) applied CNNs for automated detection, achieving high accuracy in complex terrains.
- **Integrated Approaches:** Wang et al. (2020) combined machine learning and deep learning, achieving 92.5% accuracy (ScienceDirect).
- **Our Research:** Builds on these by using U-Net for semantic segmentation, leveraging 6-channel inputs (RGB, NDVI, Slope, Elevation) and advanced preprocessing (Gaussian blur, CLAHE).

Study	Methodology	Key Findings
Koren et al. (2009)	SVM for susceptibility mapping	Effective for structured data, limited spatial analysis
Zhang & Chen (2018)	CNN for automated detection	High accuracy in complex terrains
Wang et al. (2020)	ML + DL with integrated data	92.5% accuracy in natural-terrain detection
Our Research	U-Net with 6-channel inputs	Precise pixel-wise detection, advanced preprocessing

Methodology

Data Collection and Preprocessing

- **Data Source:** High-resolution satellite imagery, labeled as "landslide" or "non-landslide." From **Kaggle**

Preprocessing:

- Resize to 128x128 pixels.
- Normalize RGB, Slope, Elevation to [0,1].
- Calculate Enhanced NDVI with Gaussian blur (5x5 kernel) and CLAHE.
- Augment with rotations and flips; replace NaN with 0.000001.
- Split: 80% training, 20% validation (801 image-mask pairs).

Methodology

Feature Extraction

- 6 Input Channels: RGB (visual), NDVI (vegetation), Slope (terrain), Elevation (altitude).
- Role: Each channel provides unique environmental insights.

Methodology

U-Net Architecture

- **U-Net:** Encoder-decoder with skip connections for semantic segmentation.
- **Details:** Input: 128x128x6; Output: 128x128x1 binary mask.
- **Parameters:** '*he_normal*' initializer, '*same*' padding.

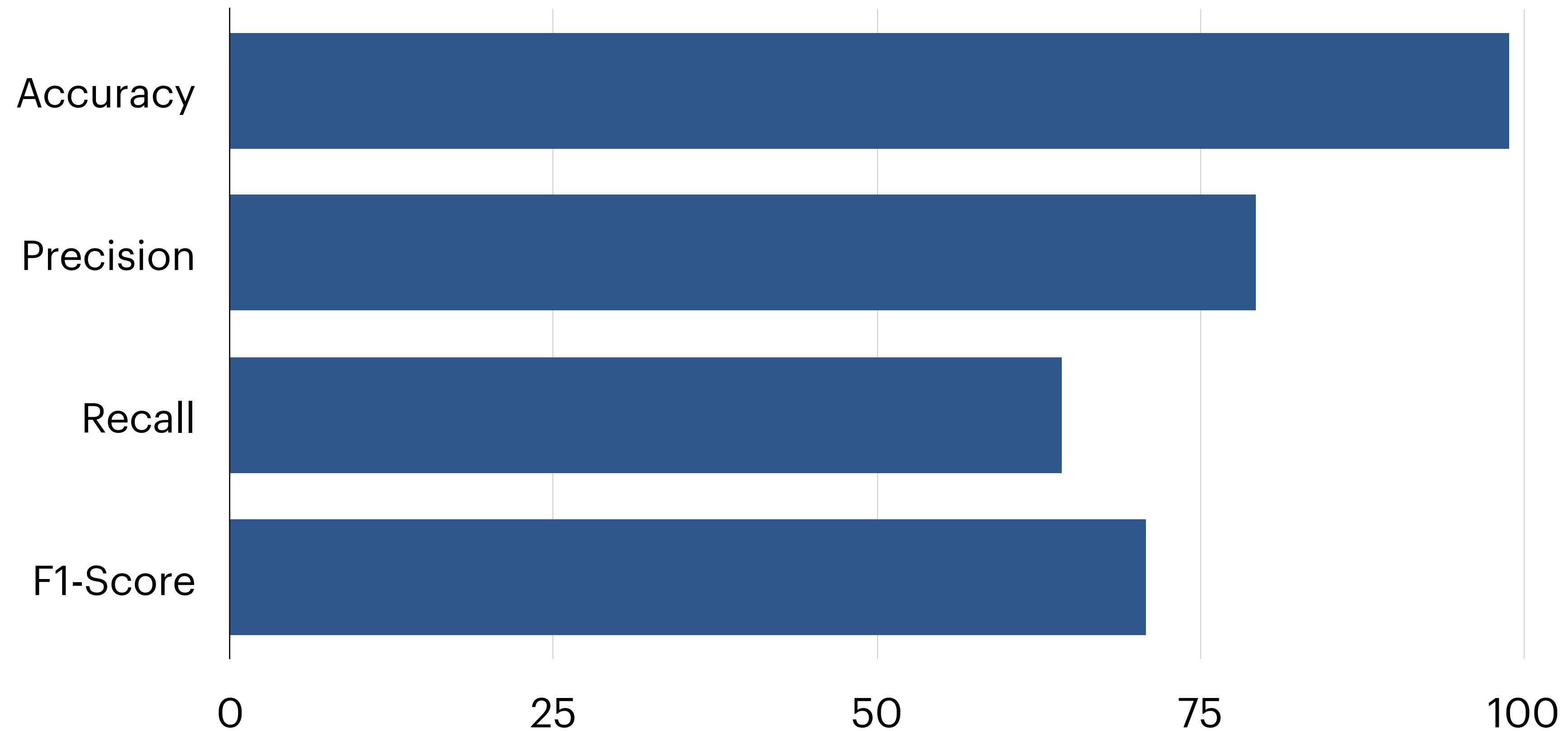
Methodology

Training Process

- Loss: Binary cross-entropy.
- Optimizer: Adam (learning rate 0.001).
- Parameters: Batch size 16, 30 epochs.
- Metrics: Accuracy, F1-score, Precision, Recall.

Results

Loss: 0.0318



Discussion

- **Interpretation:** High precision minimizes false alarms; lower recall indicates missed landslides.
- **Implications:** Enables GIS integration for real-time monitoring.
- **Challenges:** Data quality, exclusion of rainfall/soil data.
- **Future Directions:** Temporal data, hybrid models, transfer learning.

Conclusion

- U-Net achieves 98.91% accuracy for landslide detection.
- Potential for real-time monitoring and early warnings.
- Future work to improve recall and integrate multi-modal data.

Thank You!